

PayPalPayment.java - Deep Dive Explanation

1. OVERVIEW & LINKAGE

This is another Concrete Subclass. Like CreditCardPayment, it links to Payment by extending it. However, PayPal uses emails and passwords, not card numbers, proving that sibling classes can have entirely different internal data structures while sharing the same public interface.

2. DETAILED CODE BREAKDOWN

```
private String email;  
private String password;
```

Specific Encapsulated State: These fields only exist for PayPal. No card data exists here.

```
@Override  
public boolean process(...) { ... }
```

This shares the exact same method signature as CreditCardPayment.process(), but the internal algorithm is entirely different. The PaymentProcessor doesn't care **how** this works, only that it **does** work.

```
boolean validAccount = email.contains("@") && password.length() >= 4;
```

Algorithm Step 1 (Validation): It performs string analysis to ensure the email contains an "@" symbol and the password meets a minimum length requirement. If it fails, it halts and marks the status as FAILED.

```
boolean confirmed = amount <= 250_000;
```

Algorithm Step 2 (Authorization): It simulates an OAuth redirection to PayPal servers to check the user balance. It declines anything over \$250,000. If successful, it updates the inherited status field to SUCCESS.